

- 5 $\neg(p \wedge q)$ Step(1), (4) Modus Tollens
 6 $\neg p \vee \neg q$ Step (5) and (3) De morgan's Law
 The argument is valid.

Example 1.3.8

Establish the validity of the argument

$$\begin{array}{l}
 p \rightarrow q \\
 q \rightarrow (r \wedge s) \\
 \neg r \vee (\neg t \vee u) \\
 p \wedge t \\
 \dots\dots\dots \\
 \therefore u
 \end{array}$$

Solution:

	Steps	Reason
1	$p \rightarrow q$	Premises
2	$q \rightarrow (r \wedge s)$	Premises
3	$p \rightarrow (r \wedge s)$	Step (1),(2) Law of Syllogism
4	$p \wedge t$	Premises
5	p	Step (4) Rule of Conjunctive Simplification
6	$r \wedge s$	Step (5) and (3) Modus Ponens
7	r	Step (6) and rule of Conjunctive Simplification
8	$\neg r \vee (\neg t \vee u)$	Premises
9	$(\neg r \vee \neg t) \vee u$	Step (8), Associative Law
10	$\neg(r \wedge t) \vee u$	Step (9) and De Morgan's law
11	t	Step (4) Rule of Conjunctive Simplification
12	$r \wedge t$	Step (7) and (11) and the rule of Conjunction
13	u	Steps (10),(12) and Rule of disjunctive Syllogism

Example 1.3.9

Establish the validity of the argument

$$\begin{array}{l}
 p \wedge q \\
 p \rightarrow (r \wedge q) \\
 r \rightarrow (s \vee t) \\
 \neg s \\
 \dots\dots\dots \\
 \therefore t
 \end{array}$$

Solution:

	Steps	Reason
1	$p \wedge q$	Premises
2	p	Step (1) and Rule of conjunctive Simplification
3	$p \rightarrow (r \wedge q)$	Premises
4	$r \wedge q$	Step (2),(3) Modus Ponens
5	r	Step (4) Rule of conjunctive Simplification
6	$r \rightarrow (s \vee t)$	Premises
7	$s \vee t$	Step (5), (6) and Modus Ponens
8	$\neg s$	Premises
9	t	Step (7),(8) Rule of disjunctive Syllogism

Example 1.3.10

Establish the validity of the argument by the method of proof by contradiction

$$\neg p \leftrightarrow q$$

$$q \rightarrow r$$

$$\neg r$$

.....

$$\therefore p$$

Solution:

	Steps	Reason
1	$\neg p \leftrightarrow q$	Premises
2	$(\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$	Step(1)and $(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
3	$\neg p \rightarrow q$	Step (2) Rule of conjunctive Simplification
4	$q \rightarrow r$	Premises
5	$\neg p \rightarrow r$	Step (3),(4) Law of the Syllogism
6	$\neg p$	Premises (assume negation of conclusion)
7	r	Step (5), (6) and Modus Ponens
8	$\neg r$	Premises
9	$r \wedge \neg r$	Step (7), (8) Rule of Conjunction
10	F_0	Step (9) Inverse law
11	$\therefore p$	Step (6),(10) Method of proof by Contradiction

Example 1.3.11

Establish the validity of the argument by the method of proof by contradiction

$$\begin{aligned}
 &r \rightarrow \neg q \\
 &r \vee s \\
 &s \rightarrow \neg q \\
 &p \rightarrow q
 \end{aligned}$$

.....

$$\therefore \neg p$$

Solution:

	Steps	Reason
1	$r \rightarrow \neg q$	Premises
2	$s \rightarrow \neg q$	Premises
3	$(r \vee s) \rightarrow \neg q$	Step (1),(2) Rule for proof by cases
4	$r \vee s$	Premises
5	$\neg q$	Step (3), (4) Modus Ponens
6	$\neg(\neg p)$	Premises (assume negation of conclusion)
7	p	Step (6) Law of Double negation
8	$p \wedge \neg p$	Step (5), (7) Rule of Conjunction
9	F_0	Step (8) Inverse law
10	$\therefore \neg p$	Step (6),(9) Method of proof by Contradiction

Example 1.3.12

Establish the validity of the argument by the method of proof by contradiction

$$\begin{aligned}
 &p \rightarrow (q \wedge r) \\
 &\neg(q \wedge s) \\
 &r \rightarrow s
 \end{aligned}$$

.....

$$\therefore \neg p$$

Solution:

	Steps	Reason
1	$p \rightarrow (q \wedge r)$	Premises
2	$r \rightarrow s$	Premises
3	$\neg(q \wedge s)$	Premises
4	$\neg(\neg p)$	Premises (assume negation of conclusion)
5	p	Step (4) Law of Double negation
6	$q \wedge r$	Step (1), (5) Modus Ponens

7	q	Step (6) Rule of conjunctive Simplification
8	r	Step (6) Rule of conjunctive Simplification
9	s	Step (2),(8) Modus Ponens
10	$q \wedge s$	Step (7),(9) Rule of Conjunction
11	$\neg(q \wedge s) \wedge (q \wedge s)$	Step (3),(10) Rule of Conjunction
12	F_0	Step (11) Inverse law
13	$\therefore \neg p$	Step (4), (12) Method of proof by Contradiction

Example 1.3.13

Establish the validity of the argument

$$\begin{array}{l}
 u \rightarrow r \\
 (r \wedge s) \rightarrow (p \vee t) \\
 q \rightarrow (u \wedge s) \\
 \neg t \\
 \dots\dots\dots \\
 \therefore q \rightarrow p
 \end{array}$$

Solution:

	Steps	Reason
1	$u \rightarrow r$	Premises
2	$(r \wedge s) \rightarrow (p \vee t)$	Premises
3	$q \rightarrow (u \wedge s)$	Premises
4	$\neg t$	Premises
5	q	Premises (assume)
6	$u \wedge s$	Step (3), (5) Modus Ponens
7	u	Step (6) Rule of conjunctive Simplification
8	s	Step (6) Rule of conjunctive Simplification
9	r	Step (1),(7) Modus Ponens
10	$r \wedge s$	Step (8),(9) Rule of Conjunction
11	$p \vee t$	Step (2),(10) Modus Ponens
12	p	Step (11) (4) Rule of disjunctive Syllogism
13	$\therefore q \rightarrow p$	Step (5), (12)

Example 1.3.14

Establish the validity of the argument

$$p \rightarrow (q \rightarrow r)$$

$$p \vee \neg s$$

$$q$$

.....

$$\therefore s \rightarrow r$$

Solution:

	Steps	Reason
1	$p \rightarrow (q \rightarrow r)$	Premises
2	$p \vee \neg s$	Premises
3	q	Premises
4	s	Premises (assume)
5	$\neg s \vee p$	Step (2) commutative property
6	$s \rightarrow p$	Step (5) $p \rightarrow q \Leftrightarrow \neg p \vee q$
7	$s \rightarrow (q \rightarrow r)$	Step (1), (6) Law of the Syllogism
8	$\neg s \vee (q \rightarrow r)$	Step (7) $p \rightarrow q \Leftrightarrow \neg p \vee q$
9	$\neg s \vee (\neg q \vee r)$	Step (8) $p \rightarrow q \Leftrightarrow \neg p \vee q$
10	$(\neg s \vee \neg q) \vee r$	Step (9) Associative Property
11	$\neg(s \wedge q) \vee r$	Step (10) Demorgan's Law
12	$(s \wedge q) \rightarrow r$	Step (11) $p \rightarrow q \Leftrightarrow \neg p \vee q$
13	$s \wedge q$	Step (3) (4) Rule of Conjunction
14	r	Step (12), (13) Modus Ponens
15	$\therefore s \rightarrow r$	Step (4), (14)

1.4 Negation of Compound statements**Negation of conjunction**

If p and q are propositions then the negation of the conjunction $p \wedge q$ is

$$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

Negation of disjunction

If p and q are propositions then the negation of the disjunction $p \vee q$ is

$$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$$

Example 1.3.1

Rita is baking a cake. If Rita is baking a cake, then she is not practicing her flute. If Rita is not practicing her flute, then her father will not buy her a car. Therefore, Rita's father will not buy her a car. Check the validity of the statement.

Solution:

The given propositions are

p : Rita is baking a cake

q : Rita is practicing her flute

r : Rita's father will buy her a car

The argument is

p

$p \rightarrow \neg q$

$\neg q \rightarrow \neg r$

.....

$\therefore \neg r$

	Steps	Reason
1	p	Premises
2	$p \rightarrow \neg q$	Premises
3	$\neg q \rightarrow \neg r$	Premises
4	$p \rightarrow \neg r$	Step (2), (3) Law of Syllogism
5	$\neg r$	Step (1), (4) Modus Ponens

The conclusion follows from the premises. $\therefore$ the argument is valid.

Example 1.3.2

If horses or cows eat grass, then the mosquito is the national bird. If the mosquito is the national bird then peanut butter tastes good on hot dogs. But peanut butter tastes terrible on hot dogs. Therefore, the cow didn't eat grass. Establish the validity of the argument.

Solution:

The given propositions are,

p : Horses eat grass

q : Cows eat grass

r : Mosquito is the national bird

s : Peanut butter tastes good on hot dogs

The hypothesis is

$p \vee q \rightarrow r$

$r \rightarrow s$

$\neg s$

.....

conclusion is: $\therefore \neg q$

Steps		Reason
1	$p \vee q \rightarrow r$	Premises
2	$r \rightarrow s$	Premises
3	$p \vee q \rightarrow s$	Step (1), (2) Law of Syllogism
4	$\neg s$	Premises
5	$\neg(p \vee q)$	Step (3), (4) Modus Tollens
6	$\neg p \wedge \neg q$	Step 5, De morgan's Law
7	$\neg q$	Step 6, Rule of conjunctive Simplification

The conclusion follows from the premises. $\therefore$ the argument is valid.

Example 1.3.3

If I join IIT then I will get best education. If I get best education, then I will get job in USA. If I get job in USA then I will become a millionaire. Therefore, if I joined in IIT then I will become a millionaire. Show that the conclusion follows from the premises.

Solution:

The given propositions are

p : I join IIT

q : I will get best education

r : I will get job in USA

s : I become a millionaire

The argument is $p \rightarrow q$

$q \rightarrow r$

$r \rightarrow s$

.....

$\therefore p \rightarrow s$

Steps	Reason
1 $p \rightarrow q$	Premises
2 $q \rightarrow r$	Premises
3 $p \rightarrow r$	Step (1),(2) Law of Syllogism
4 $r \rightarrow s$	Premises
5 $p \rightarrow s$	Step (4) Law of Syllogism

$\therefore$ the conclusion follows from the premises.

Example 1.3.4

If the band could not play rock music or the refreshments were not delivered on time, then the new year party would have been cancelled and Alice would have been angry. If the party were cancelled, then refunds had to be made. No refunds were made. Therefore, the band could play rock music. Establish the validity of the statement.

Solution:

The given propositions are,

p : The band play rock music

q : The refreshments were delivered on time

r : The new year party was cancelled

s : Alice was angry

t : Refunds were made

The given argument is,

$(\neg p \vee \neg q) \rightarrow (r \wedge s)$

$r \rightarrow t$

$\neg t$

.....

$\therefore p$

	Steps	Reason
1	$r \rightarrow t$	Premises
2	$\neg t$	Premises
3	$\neg r$	Step (1),(2) Modus Tollens
4	$\neg r \vee \neg s$	Step(3) Rule of disjunctive Amplification
5	$\neg(r \wedge s)$	Step (4) De morgan's Law
6	$(\neg p \vee \neg q) \rightarrow (r \wedge s)$	Premises
7	$\neg(\neg p \vee \neg q)$	Step (5),(6), Modus Tollens
8	$p \wedge q$	Step (7), Demorgan's Law and law of double negation
9	p	Step (8), Rule of conjunctive Simplification

The argument is valid since the conclusion follows from the premises.

Example 1.3.5

Let p, q, r be the statements given as

p : Arjun studies. q : He plays cricket. r : He passes Data Structures.

Let p_1, p_2, p_3 denote the premises.

p_1 : If Arjun studies, then he will pass Data Structures.

p_2 : If he doesn't play cricket, then he will study.

p_3 : He failed Data Structures.

Determine whether the argument $(p_1 \wedge p_2 \wedge p_3) \rightarrow q$ is valid.

Solution:

The argument is,

$$p_1: p \rightarrow r$$

$$p_2: \neg q \rightarrow p$$

$$p_3: \neg r$$

.....

$$\therefore q$$

	Steps	Reason
1	$p \rightarrow r$	Premises
2	$\neg q \rightarrow p$	Premises

3	$\neg r$	Premises
4	$\neg q \rightarrow r$	Step (1) (2) Law of Syllogism
5	$\neg(\neg q)$	Step (3), (4) Modus Tollens
6	q	Step (5) Law of double negation

The argument is valid since the conclusion follows from the premises.

Example 1.3.6

If Dominic goes to the racetrack, then Helen will be mad. If Ralph plays cards all night, then Carmela will be mad. If either Helen or Carmela gets mad, then Veronica will be notified. Veronica has not heard from either of these clients, that is Veronica is not notified. Consequently, Dominic did not make it to the racetrack and Ralph didn't play cards all night. Establish the validity of the statement.

Solution:

The given propositions are

p : Dominic goes to the racetrack

q : Helen is mad

r : Ralph plays cards

s : Carmela is mad

t : Veronica will be notified

Then the hypotheses are

$$p \rightarrow q$$

$$r \rightarrow s$$

$$q \vee s \rightarrow t$$

$$\neg t$$

.....

$$\therefore \neg p \wedge \neg r$$

Steps	Reason
1 $q \vee s \rightarrow t$	Premises
2 $\neg t$	Premises
3 $\neg(q \vee s)$	Step (1),(2) Modus Tollens
4 $\neg q \wedge \neg s$	Step 3 , De Morgan's Law

5	$\neg q$	Step (4) Rule of conjunctive Simplification
6	$\neg s$	Step (4) Rule of conjunctive Simplification
7	$p \rightarrow q$	Premises
8	$\neg p$	Step (5),(7) Modus Tollens
9	$r \rightarrow s$	Premises
10	$\neg r$	Step (6),(9) Modus Tollens
11	$\neg p \wedge \neg r$	Step (8), (10) Rule of Conjunction

The argument is valid.

Example 1.3.7

If Rachel gets the supervisor's position and works hard, then she will get a raise. If she gets the raise, then she will buy a new car. She has not purchased a new car. Therefore, either Rachel did not get the supervisor's position or she did not work hard. Show that the conclusion follows from the premises.

Solution:

The given propositions are

p : Rachel gets the supervisor's position

q : Rachel works hard

r : Rachel gets the raise

s : Rachel buy a new car

The argument is

$$(p \wedge q) \rightarrow r$$

$$r \rightarrow s$$

$$\neg s$$

.....

$$\therefore \neg p \vee \neg q$$

Steps		Reason
1	$(p \wedge q) \rightarrow r$	Premises
2	$r \rightarrow s$	Premises
3	$\neg s$	Premises
4	$\neg r$	Step (3), (4) Modus Tollens

- 5 $\neg(p \wedge q)$ Step(1), (4) Modus Tollens
 6 $\neg p \vee \neg q$ Step (5) and (3) De morgan's Law
 The argument is valid.

Example 1.3.8

Establish the validity of the argument

$$\begin{array}{l}
 p \rightarrow q \\
 q \rightarrow (r \wedge s) \\
 \neg r \vee (\neg t \vee u) \\
 p \wedge t \\
 \dots\dots\dots \\
 \therefore u
 \end{array}$$

Solution:

Steps	Reason
1 $p \rightarrow q$	Premises
2 $q \rightarrow (r \wedge s)$	Premises
3 $p \rightarrow (r \wedge s)$	Step (1),(2) Law of Syllogism
4 $p \wedge t$	Premises
5 p	Step (4) Rule of Conjunctive Simplification
6 $r \wedge s$	Step (5) and (3) Modus Ponens
7 r	Step (6) and rule of Conjunctive Simplification
8 $\neg r \vee (\neg t \vee u)$	Premises
9 $(\neg r \vee \neg t) \vee u$	Step (8), Associative Law
10 $\neg(r \wedge t) \vee u$	Step (9) and De Morgan's law
11 t	Step (4) Rule of Conjunctive Simplification
12 $r \wedge t$	Step (7) and (11) and the rule of Conjunction
13 u	Steps (10),(12) and Rule of disjunctive Syllogism

Example 1.3.9

Establish the validity of the argument

$$\begin{array}{l}
 p \wedge q \\
 p \rightarrow (r \wedge q) \\
 r \rightarrow (s \vee t) \\
 \neg s \\
 \dots\dots\dots \\
 \therefore t
 \end{array}$$

Solution:

	Steps	Reason
1	$p \wedge q$	Premises
2	p	Step (1) and Rule of conjunctive Simplification
3	$p \rightarrow (r \wedge q)$	Premises
4	$r \wedge q$	Step (2),(3) Modus Ponens
5	r	Step (4) Rule of conjunctive Simplification
6	$r \rightarrow (s \vee t)$	Premises
7	$s \vee t$	Step (5), (6) and Modus Ponens
8	$\neg s$	Premises
9	t	Step (7),(8) Rule of disjunctive Syllogism

Example 1.3.10

Establish the validity of the argument by the method of proof by contradiction

$$\begin{array}{c}
 \neg p \leftrightarrow q \\
 q \rightarrow r \\
 \neg r \\
 \dots\dots\dots \\
 \therefore p
 \end{array}$$

Solution:

	Steps	Reason
1	$\neg p \leftrightarrow q$	Premises
2	$(\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$	Step(1)and $(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
3	$\neg p \rightarrow q$	Step (2) Rule of conjunctive Simplification
4	$q \rightarrow r$	Premises
5	$\neg p \rightarrow r$	Step (3),(4) Law of the Syllogism
6	$\neg p$	Premises (assume negation of conclusion)
7	r	Step (5), (6) and Modus Ponens
8	$\neg r$	Premises
9	$r \wedge \neg r$	Step (7), (8) Rule of Conjunction
10	F_0	Step (9) Inverse law
11	$\therefore p$	Step (6),(10) Method of proof by Contradiction

Example 1.3.11

Establish the validity of the argument by the method of proof by contradiction